

DEPARTMENT OF MATHEMATICS AND COMPUTING
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

Programme: B.Tech. Semester-IV
Paper Code: MAIC 204
Course Title: Applied Linear Algebra
Date: 08-05-2026

Examination: Mid Term II
Paper: REGULAR and REAPPEAR
M.M.: 15
Time: 11:00 AM – 11:50 AM

Note: Attempt all the Questions. Each question carries equal marks.

1. Solve by LU decomposition method:

$$x_1 + x_2 + x_3 = 2,$$

$$4x_1 + 3x_2 - x_3 = 10,$$

$$3x_1 + 5x_2 + 3x_3 = 7.$$

2. Determine all the possible Jordan Canonical forms of the linear operator $T: V \rightarrow V$ whose characteristic polynomial is $\Delta(t) = (t - 2)^3(t - 5)^2$. In each case, find the minimal polynomial $m(t)$.
3. Let V be an inner product space and let $\beta_1, \dots, \beta_n$ be any independent vectors in V . Then prove that one may construct orthogonal vectors $\alpha_1, \dots, \alpha_n$ in V such that for each $k = 1, 2, \dots, n$ the set $\{\alpha_1, \dots, \alpha_k\}$ is basis for the subspace spanned by $\beta_1, \dots, \beta_k$.